

Statistical Data Analysis

Lecture 10: Analysis of categorical data

Marc Corstanje

Periods 4 & 5

⏮ Chapter 6: Nonparametric tests

💬 Still today: discussion of Assignment 5 & Quiz 5

📍 Chapter 7: Analysis of categorical data

- Section 7.1 – Fisher's exact test for 2×2 tables
- Section 7.2 – The chi-square test for contingency tables
- Section 7.3 – Identification of cells with extreme values
- ~~Section 7.4 – The bootstrap for contingency tables~~

⏭ Chapter 8: Linear regression analysis

Categorical Data & Contingency Tables

Categorical Data & Contingency Tables

- ② What are **categorical** variables?
 - › A categorical variable splits the data into categories
 - › Example: Blood type: *O*, *A*, *B*, *AB*.
- › The categories of a categorical variable are called **levels**
- › We are often interested in relationships between two (or more) categorical variables **A** and **B**.
 - ② Are they independent or not?

Categorical Data & Contingency Tables

> $r \times c$ contingency table

- > n subjects are divided into the categories of two variables, A and B
- > N_{ij} is the number of subjects (count) in cell (i, j)
- > $N_{i.}$ and $N_{.j}$ are the **marginals** (row resp. column sums)

	B_1		B_j	...	B_c	TOT
A_1	N_{11}		N_{1j}	...	N_{1c}	$N_{1.}$
⋮	⋮		⋮		⋮	⋮
A_i	N_{i1}		N_{ij}	...	N_{ic}	$N_{i.}$
⋮	⋮		⋮		⋮	⋮
A_r	N_{r1}		N_{rj}	...	N_{rc}	$N_{r.}$
TOT	$N_{.1}$		$N_{.j}$	...	$N_{.c}$	$n = N_{..}$

- > **Goal:** investigate the relationship between row and column variable

Fisher's Exact Test for 2×2 Tables

Fisher's Exact Test for 2×2 Tables – Idea

- Are study programme and operating system choice independent?

			TOT
Math	3	2	$N_{1\cdot} = 5$
BA	7	11	$N_{2\cdot} = 18$
TOT	$N_{\cdot 1} = 10$	$N_{\cdot 2} = 13$	$n = 23$

- Null hypothesis: independence
- For **fixed (!) marginals**, table determined by $N_{11} \rightsquigarrow$ use as **test statistic**
- Conditional on marginals**, the **null** distribution of N_{11} is **hypergeometric**

$$\mathbb{P}_{H_0}(N_{11} = k) = \frac{\binom{N_{1\cdot}}{k} \binom{n - N_{1\cdot}}{N_{\cdot 1} - k}}{\binom{n}{N_{\cdot 1}}} \rightsquigarrow \text{e.g. } \mathbb{P}_{H_0}(N_{11} = 3) = \frac{\binom{5}{3} \binom{18}{7}}{\binom{23}{10}} \approx 0.278$$

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- Note that  uses somewhat conflicting notation, see also "dhyper"

```
# x = nr. of white balls drawn (N_11)
# m = nr. of white balls in urn (N_1.)
# n = nr. of black balls in urn (N_2. = n - N_1.)
# k = nr. of drawn balls (N_.1)
#
dhyper(x = 3, m = 5, n = 18, k = 10)
```

```
## [1] 0.2781658
```

	Drawn	Not Drawn	TOT
White Balls	3	2	5
Black Balls	7	11	18
TOT	10	13	23

Fisher's Exact Test for 2×2 Tables – Illustration

- > What is the p -value for the two-sided test (H_0 : independence)?
 - > Recall: probability of an **at least as extreme** result under H_0
 - > **Intuition**: take all possible tables with given marginals and varying N_{11} , calculate the corresponding null probabilities and sum those that are **smaller or equal** than that of the **original** table

	☐	🍏	TOT
Math	5	0	5
BA	5	13	18
TOT	10	13	23

$\mathbb{P}_{H_0}(N_{11} = 5) \approx 0.007$

	☐	🍏	TOT
Math	4	1	5
BA	6	12	18
TOT	10	13	23

$\mathbb{P}_{H_0}(N_{11} = 4) \approx 0.081$

	☐	🍏	TOT
Math	3	2	5
BA	7	11	18
TOT	10	13	23

$\mathbb{P}_{H_0}(N_{11} = 3) \approx 0.278$

	☐	🍏	TOT
Math	2	3	5
BA	8	10	18
TOT	10	13	23

$\mathbb{P}_{H_0}(N_{11} = 2) \approx 0.382$

	☐	🍏	TOT
Math	1	4	5
BA	9	9	18
TOT	10	13	23

$\mathbb{P}_{H_0}(N_{11} = 1) \approx 0.212$

	☐	🍏	TOT
Math	0	5	5
BA	10	8	18
TOT	10	13	23

$\mathbb{P}_{H_0}(N_{11} = 0) \approx 0.038$

$$p\text{-value} \approx 0.007 + 0.081 + 0.278 + 0.212 + 0.038 = 0.616 > 0.05 = \alpha$$

Fisher's Exact Test for 2×2 Tables – R Output

➤ In R, we can use `fisher.test`

```
(OS_choices <- matrix(c(3, 2, 7, 11), byrow = TRUE, nrow = 2,
  dimnames = list(Programme = c("Math", "BA"), OS = c("Windows", "macOS"))))

##           OS
## Programme Windows macOS
##      Math      3      2
##      BA       7     11

fisher.test(OS_choices, conf.int = FALSE)

##
##  Fisher's Exact Test for Count Data
##
## data:  OS_choices
## p-value = 0.6175
## alternative hypothesis: true odds ratio is not equal to 1
## sample estimates:
## odds ratio
##    2.268156
```

➤ For 2×2 tables, the null of independence is **equivalent** to the null hypothesis that the following **odds ratio** is one

$$\frac{\mathbb{P}(\text{OS} = \blacksquare | \text{BSc} = \text{Math}) / \mathbb{P}(\text{OS} = \blacktriangle | \text{BSc} = \text{Math})}{\mathbb{P}(\text{OS} = \blacksquare | \text{BSc} = \text{BA}) / \mathbb{P}(\text{OS} = \blacktriangle | \text{BSc} = \text{BA})} = \frac{\mathbb{P}(\text{OS} = \blacksquare | \text{BSc} = \text{Math}) / (1 - \mathbb{P}(\text{OS} = \blacksquare | \text{BSc} = \text{Math}))}{\mathbb{P}(\text{OS} = \blacksquare | \text{BSc} = \text{BA}) / (1 - \mathbb{P}(\text{OS} = \blacksquare | \text{BSc} = \text{BA}))}$$

❓ For 2×2 tables, how can one state a one-sided alternative?

Chi-Square Test for $r \times c$ Tables

Chi-Square Test for $r \times c$ Tables – Idea

- What if we have a table larger than 2×2 ?

	Ulcer	Cancer	Control	TOT
O	983	383	2892	4258
A	679	416	2625	3720
B	134	84	570	788
TOT	1796	883	6087	8766

- For tables larger than 2×2 , Fisher's exact test does **not** apply
- We can instead use an approximate **chi-square test** (a.k.a. χ^2 test)

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- > For tables larger than 2×2 , Fisher's exact test does **not** apply
- > We can instead use an approximate **chi-square test** (a.k.a. χ^2 test)
- > **Intuition:** 'one-way' ($1 \times k$) table $\rightsquigarrow$ **multinomial** experiment

	Cat. 1	...	Cat. k	TOT
Count	m_1	...	m_k	m

- > Vector of counts $(m_1, \dots, m_k) \sim \text{Multin}(m, p_1, \dots, p_k)$
- > Example: roll a (fair) dice m times, $(\#1s, \dots, \#6s) \sim \text{Multin}(m, 1/6, \dots, 1/6)$
- > **Theorem 1** in Syllabus: $\sum_{j=1}^k \frac{(m_j - mp_j)^2}{mp_j} \xrightarrow{\mathcal{D}} \chi_{k-1}^2$ as $m \rightarrow \infty$
- > **Extension:** model for $r \times c$ table $\rightsquigarrow$ set of multinomial r.v.s
 - > **Theorem 2** in Syllabus: **sum** of s independent χ_{ν}^2 -distributed r.v.s $\sim \chi_{s\nu}^2$

Chi-Square Test for $r \times c$ Tables – Independence vs. Homogeneity

- Test statistic χ^2 compares observed and expected (under H_0) counts



- Null of independence between row and column variable (Model A)
 - Null of homogeneity: distribution of given categorical variable is the same in each subgroup
 - Model B considers distribution of column variable in each row
 - Model C considers distribution of row variable in each column
- ② Assume the blood group proportions in a given population are O : 45%, A : 43% and B = 9%. You draw a single random sample of $n = 300$ people from this population, and classify them as before. Which model specification does this situation fall under? What can you say about the row sums?
- ② Assume the same situation, except now you draw three independent random samples (one for each blood group) of size 100. How would your previous answers change?

Chi-Square Test for $r \times c$ Tables – Model Comparison

	Model A	Model B	Model C
# Independent Samples	1	r	c
Sample Size(s)	n	$N_{i\cdot}, i = 1, \dots, r$	$N_{\cdot j}, j = 1, \dots, c$
Stochastic Margins?	All margins	Column margins	Row margins
Null Hypothesis	$p_{ij} = p_{i\cdot} \cdot p_{\cdot j}$	$p_{1j} = p_{2j} = \dots = p_{rj} \equiv p_j$	$p_{i1} = p_{i2} = \dots = p_{ic} \equiv p_i$

- › Row indices: $i = 1, \dots, r$, column indices $j = 1, \dots, c$
- › Same test statistic: $X^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(N_{ij} - n\hat{p}_{ij})^2}{n\hat{p}_{ij}}$, where $p_{ij} = \frac{N_{i\cdot}}{n} \times \frac{N_{\cdot j}}{n}$
- › Same approximate null distribution for large n and large row- and column-sums: $X^2 \sim \chi^2_{(r-1)(c-1)}$
 - › **Caveat:** this approximation is reasonable if (rule-of-thumb) $\mathbb{E}_{H_0}(N_{ij}) > 1$ for all (i, j) and $\mathbb{E}_{H_0}(N_{ij}) > 5$ for at least 80% of the cells
 - › If the rule is violated, one can use the **bootstrap**

Chi-Square Test for 2×2 Tables – R Outputs

```
### R default (note continuity correction for 2 x 2 tables)
chisq.test(OS_choices)

##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: OS_choices
## X-squared = 0.11058, df = 1, p-value = 0.7395

### Without continuity correction
chisq.test(OS_choices, correct = FALSE)

##
## Pearson's Chi-squared test
##
## data: OS_choices
## X-squared = 0.70966, df = 1, p-value = 0.3996

### Rule of thumb is violated -> continuity correction useful

### Using the bootstrap (function argument 'simulate.p.value')
# Setting a seed is good practice in this case, as there is randomness involved
set.seed(123)
chisq.test(OS_choices, simulate.p.value = TRUE)

##
## Pearson's Chi-squared test with simulated p-value (based on 2000
## replicates)
##
## data: OS_choices
## X-squared = 0.70966, df = NA, p-value = 0.6247
```


Extremes

Extremes – Determination of Outliers

- If H_0 **rejected**, which categories of the two variables are **involved** in the relationship?
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- › **Contributions:** $\frac{N_{ij} - n\hat{p}_{ij}}{\sqrt{n\hat{p}_{ij}}}$
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 - › In **R**: `residuals`
 - › This is one of several normalization options
- › **Normalized residuals**: $V_{ij} = \frac{N_{ij} - n\hat{p}_{ij}}{\sqrt{\frac{N_{i\cdot}(n - N_{i\cdot})N_{\cdot j}(n - N_{\cdot j})}{n^2(n-1)}}} \stackrel{H_0}{\sim} N(0, 1)$ (approx.)
 - › In **R**: `stdres`
 - › This is another normalization option
 - › The approximate normal distribution of these residuals is useful in quantifying 'extremeness'
 - › Their **empirical** distribution can also be used

```
chisq.test(diseases) # H_0 rejected
```

```
##  
## Pearson's Chi-squared test  
##  
## data: diseases  
## X-squared = 40.543, df = 4, p-value = 3.341e-08
```

```
diseases - chisq.test(diseases)$expected # "Residuals" in Syllabus
```

```
##           Disease  
## Blood Group   Ulcer      Cancer   Control  
##           0 110.61031 -45.908738 -64.70157  
##           A -83.16290  41.284052  41.87885  
##           B -27.44741   4.624686  22.82272
```

```
chisq.test(diseases)$residuals # "Contributions" in Syllabus
```

```
##           Disease  
## Blood Group   Ulcer      Cancer   Control  
##           0  3.744901 -2.2167306 -1.1899017  
##           A -3.012352  2.1327072  0.8239908  
##           B -2.160160  0.5190863  0.9756711
```

```
chisq.test(diseases)$stdres # Normalized residuals
```

```
##           Disease  
## Blood Group   Ulcer      Cancer   Control  
##           0  5.856437 -3.2596927 -3.001472  
##           A -4.452636  2.9642396  1.964553  
##           B -2.539358  0.5737837  1.849999
```

```
# Compare with e.g. the 2.5% and 97.5% quantiles of  $N(0, 1)$ , i.e.  $\pm 1.96$   
# For example, (0, Ulcer) seems to occur more frequently than expected under  $H_0$ 
```

Summary and wrap-up

- › Categorical variables split data into categories
 - › Use contingency tables to summarise the division of the data
 - › Often we test for independence or homogeneity
- › Fisher's Exact test
 - › For 2×2 contingency tables (variables with 2 levels)
 - › Under H_0 , $N_{11} \sim \text{hypergeom.}$
 - › Compare odds ratios
- › χ^2 -test for $r \times c$ contingency tables
 - › Compare expected numbers in tables to actual numbers
 - › Null distribution is χ^2
 - › Test for homogeneity or independence
- › Consider Extremes to identify which categories cause rejection of H_0
 - › Residuals
 - › Contributions
 - › Normalized residuals

What now?

› To do for now

- › Start with assignment 6
- › Deadline: May 6 (!) at 23:59
- › All material for submission assignment covered, not everything for the quiz.
- › Good luck!

› Next week

- › Start linear regression analysis